NCERT CLASS 9
60 Questions Per Topic

Based on the refined 2026–27 curriculum map created for this project.

Each curriculum topic below contains exactly 60 questions: understanding, examples/classification, comparison, application, reasoning/error analysis, and higher-order activity.

These are question-bank items derived from the topic descriptions in the supplied curriculum map. Validate against the exact school/textbook edition before high-stakes use.

MATHEMATICS — 1. Number Systems

Coverage: ■ Real numbers; rational/irrational numbers; representations; exponents; approximations.

Understanding

1. What is ■ Real numbers, and what are its main features? Use a simple example.

2. What is rational/irrational numbers, and what are its main features? Use a school-life example.

3. What is representations, and what are its main features? Use a real-world example.

4. What is exponents, and what are its main features? Show the idea with a diagram where appropriate.

5. What is approximations., and what are its main features? Explain in your own words.

6. What is ■ Real numbers, and what are its main features? Give a step-by-step response.

7. What is rational/irrational numbers, and what are its main features? Mention one common misconception.

8. What is representations, and what are its main features? State one practical application.

9. What is exponents, and what are its main features? Justify your response.

10. What is approximations., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Real numbers and explain each. Use a simple example.

12. Give two examples related to rational/irrational numbers and explain each. Use a school-life example.

13. Give two examples related to representations and explain each. Use a real-world example.

14. Give two examples related to exponents and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to approximations. and explain each. Explain in your own words.

16. Give two examples related to ■ Real numbers and explain each. Give a step-by-step response.

17. Give two examples related to rational/irrational numbers and explain each. Mention one common misconception.

18. Give two examples related to representations and explain each. State one practical application.

19. Give two examples related to exponents and explain each. Justify your response.

20. Give two examples related to approximations. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Real numbers related to rational/irrational numbers? Compare the two ideas. Use a simple example.

22. How is rational/irrational numbers related to representations? Compare the two ideas. Use a school-life example.

23. How is representations related to exponents? Compare the two ideas. Use a real-world example.

24. How is exponents related to approximations.? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is approximations. related to ■ Real numbers? Compare the two ideas. Explain in your own words.

26. How is ■ Real numbers related to rational/irrational numbers? Compare the two ideas. Give a step-by-step response.

27. How is rational/irrational numbers related to representations? Compare the two ideas. Mention one common misconception.

28. How is representations related to exponents? Compare the two ideas. State one practical application.

29. How is exponents related to approximations.? Compare the two ideas. Justify your response.

30. How is approximations. related to ■ Real numbers? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Real numbers to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply rational/irrational numbers to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply representations to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply exponents to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply approximations. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Real numbers to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply rational/irrational numbers to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply representations to a realistic situation. What would you do or calculate? State one practical application.

39. Apply exponents to a realistic situation. What would you do or calculate? Justify your response.

40. Apply approximations. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Real numbers. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of rational/irrational numbers. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of representations. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of exponents. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of approximations.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Real numbers. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of rational/irrational numbers. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of representations. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of exponents. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of approximations.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Real numbers. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show rational/irrational numbers. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show representations. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show exponents. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show approximations.. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Real numbers. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show rational/irrational numbers. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show representations. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show exponents. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show approximations.. Include the relevant rule or principle.

MATHEMATICS — 2. Polynomials

Coverage: ■ Terms/degree; zeroes; identities; factorisation; algebraic reasoning.

Understanding

1. What is ■ Terms/degree, and what are its main features? Use a simple example.

2. What is zeroes, and what are its main features? Use a school-life example.

3. What is identities, and what are its main features? Use a real-world example.

4. What is factorisation, and what are its main features? Show the idea with a diagram where appropriate.

5. What is algebraic reasoning., and what are its main features? Explain in your own words.

6. What is ■ Terms/degree, and what are its main features? Give a step-by-step response.

7. What is zeroes, and what are its main features? Mention one common misconception.

8. What is identities, and what are its main features? State one practical application.

9. What is factorisation, and what are its main features? Justify your response.

10. What is algebraic reasoning., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Terms/degree and explain each. Use a simple example.

12. Give two examples related to zeroes and explain each. Use a school-life example.

13. Give two examples related to identities and explain each. Use a real-world example.

14. Give two examples related to factorisation and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to algebraic reasoning. and explain each. Explain in your own words.

16. Give two examples related to ■ Terms/degree and explain each. Give a step-by-step response.

17. Give two examples related to zeroes and explain each. Mention one common misconception.

18. Give two examples related to identities and explain each. State one practical application.

19. Give two examples related to factorisation and explain each. Justify your response.

20. Give two examples related to algebraic reasoning. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Terms/degree related to zeroes? Compare the two ideas. Use a simple example.

22. How is zeroes related to identities? Compare the two ideas. Use a school-life example.

23. How is identities related to factorisation? Compare the two ideas. Use a real-world example.

24. How is factorisation related to algebraic reasoning.? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is algebraic reasoning. related to ■ Terms/degree? Compare the two ideas. Explain in your own words.

26. How is ■ Terms/degree related to zeroes? Compare the two ideas. Give a step-by-step response.

27. How is zeroes related to identities? Compare the two ideas. Mention one common misconception.

28. How is identities related to factorisation? Compare the two ideas. State one practical application.

29. How is factorisation related to algebraic reasoning.? Compare the two ideas. Justify your response.

30. How is algebraic reasoning. related to ■ Terms/degree? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Terms/degree to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply zeroes to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply identities to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply factorisation to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply algebraic reasoning. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Terms/degree to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply zeroes to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply identities to a realistic situation. What would you do or calculate? State one practical application.

39. Apply factorisation to a realistic situation. What would you do or calculate? Justify your response.

40. Apply algebraic reasoning. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Terms/degree. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of zeroes. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of identities. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of factorisation. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of algebraic reasoning.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Terms/degree. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of zeroes. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of identities. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of factorisation. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of algebraic reasoning.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Terms/degree. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show zeroes. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show identities. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show factorisation. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show algebraic reasoning.. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Terms/degree. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show zeroes. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show identities. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show factorisation. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show algebraic reasoning.. Include the relevant rule or principle.

MATHEMATICS — 3. Coordinate Geometry

Coverage: ■ Cartesian plane; axes; quadrants; plotting and interpreting points.

Understanding

1. What is ■ Cartesian plane, and what are its main features? Use a simple example.

2. What is axes, and what are its main features? Use a school-life example.

3. What is quadrants, and what are its main features? Use a real-world example.

4. What is plotting and interpreting points., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Cartesian plane, and what are its main features? Explain in your own words.

6. What is axes, and what are its main features? Give a step-by-step response.

7. What is quadrants, and what are its main features? Mention one common misconception.

8. What is plotting and interpreting points., and what are its main features? State one practical application.

9. What is ■ Cartesian plane, and what are its main features? Justify your response.

10. What is axes, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to quadrants and explain each. Use a simple example.

12. Give two examples related to plotting and interpreting points. and explain each. Use a school-life example.

13. Give two examples related to ■ Cartesian plane and explain each. Use a real-world example.

14. Give two examples related to axes and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to quadrants and explain each. Explain in your own words.

16. Give two examples related to plotting and interpreting points. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Cartesian plane and explain each. Mention one common misconception.

18. Give two examples related to axes and explain each. State one practical application.

19. Give two examples related to quadrants and explain each. Justify your response.

20. Give two examples related to plotting and interpreting points. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Cartesian plane related to axes? Compare the two ideas. Use a simple example.

22. How is axes related to quadrants? Compare the two ideas. Use a school-life example.

23. How is quadrants related to plotting and interpreting points.? Compare the two ideas. Use a real-world example.

24. How is plotting and interpreting points. related to ■ Cartesian plane? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Cartesian plane related to axes? Compare the two ideas. Explain in your own words.

26. How is axes related to quadrants? Compare the two ideas. Give a step-by-step response.

27. How is quadrants related to plotting and interpreting points.? Compare the two ideas. Mention one common misconception.

28. How is plotting and interpreting points. related to ■ Cartesian plane? Compare the two ideas. State one practical application.

29. How is ■ Cartesian plane related to axes? Compare the two ideas. Justify your response.

30. How is axes related to quadrants? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply quadrants to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply plotting and interpreting points. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Cartesian plane to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply axes to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply quadrants to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply plotting and interpreting points. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Cartesian plane to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply axes to a realistic situation. What would you do or calculate? State one practical application.

39. Apply quadrants to a realistic situation. What would you do or calculate? Justify your response.

40. Apply plotting and interpreting points. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Cartesian plane. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of axes. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of quadrants. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of plotting and interpreting points.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Cartesian plane. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of axes. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of quadrants. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of plotting and interpreting points.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Cartesian plane. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of axes. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show quadrants. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show plotting and interpreting points.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Cartesian plane. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show axes. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show quadrants. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show plotting and interpreting points.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Cartesian plane. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show axes. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show quadrants. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show plotting and interpreting points.. Include the relevant rule or principle.

MATHEMATICS — 4. Linear Equations in Two Variables

Coverage: ■ Solutions; tables; graphs; interpretation of relationships.

Understanding

1. What is ■ Solutions, and what are its main features? Use a simple example.

2. What is tables, and what are its main features? Use a school-life example.

3. What is graphs, and what are its main features? Use a real-world example.

4. What is interpretation of relationships., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Solutions, and what are its main features? Explain in your own words.

6. What is tables, and what are its main features? Give a step-by-step response.

7. What is graphs, and what are its main features? Mention one common misconception.

8. What is interpretation of relationships., and what are its main features? State one practical application.

9. What is ■ Solutions, and what are its main features? Justify your response.

10. What is tables, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to graphs and explain each. Use a simple example.

12. Give two examples related to interpretation of relationships. and explain each. Use a school-life example.

13. Give two examples related to ■ Solutions and explain each. Use a real-world example.

14. Give two examples related to tables and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to graphs and explain each. Explain in your own words.

16. Give two examples related to interpretation of relationships. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Solutions and explain each. Mention one common misconception.

18. Give two examples related to tables and explain each. State one practical application.

19. Give two examples related to graphs and explain each. Justify your response.

20. Give two examples related to interpretation of relationships. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Solutions related to tables? Compare the two ideas. Use a simple example.

22. How is tables related to graphs? Compare the two ideas. Use a school-life example.

23. How is graphs related to interpretation of relationships.? Compare the two ideas. Use a real-world example.

24. How is interpretation of relationships. related to ■ Solutions? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Solutions related to tables? Compare the two ideas. Explain in your own words.

26. How is tables related to graphs? Compare the two ideas. Give a step-by-step response.

27. How is graphs related to interpretation of relationships.? Compare the two ideas. Mention one common misconception.

28. How is interpretation of relationships. related to ■ Solutions? Compare the two ideas. State one practical application.

29. How is ■ Solutions related to tables? Compare the two ideas. Justify your response.

30. How is tables related to graphs? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply graphs to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply interpretation of relationships. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Solutions to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply tables to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply graphs to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply interpretation of relationships. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Solutions to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply tables to a realistic situation. What would you do or calculate? State one practical application.

39. Apply graphs to a realistic situation. What would you do or calculate? Justify your response.

40. Apply interpretation of relationships. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Solutions. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of tables. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of graphs. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of interpretation of relationships.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Solutions. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of tables. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of graphs. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of interpretation of relationships.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Solutions. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of tables. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show graphs. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show interpretation of relationships.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Solutions. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show tables. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show graphs. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show interpretation of relationships.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Solutions. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show tables. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show graphs. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show interpretation of relationships.. Include the relevant rule or principle.

MATHEMATICS — 5. Introduction to Euclid’s Geometry

Coverage: ■ Definitions; axioms; postulates; deductive reasoning.

Understanding

1. What is ■ Definitions, and what are its main features? Use a simple example.

2. What is axioms, and what are its main features? Use a school-life example.

3. What is postulates, and what are its main features? Use a real-world example.

4. What is deductive reasoning., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Definitions, and what are its main features? Explain in your own words.

6. What is axioms, and what are its main features? Give a step-by-step response.

7. What is postulates, and what are its main features? Mention one common misconception.

8. What is deductive reasoning., and what are its main features? State one practical application.

9. What is ■ Definitions, and what are its main features? Justify your response.

10. What is axioms, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to postulates and explain each. Use a simple example.

12. Give two examples related to deductive reasoning. and explain each. Use a school-life example.

13. Give two examples related to ■ Definitions and explain each. Use a real-world example.

14. Give two examples related to axioms and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to postulates and explain each. Explain in your own words.

16. Give two examples related to deductive reasoning. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Definitions and explain each. Mention one common misconception.

18. Give two examples related to axioms and explain each. State one practical application.

19. Give two examples related to postulates and explain each. Justify your response.

20. Give two examples related to deductive reasoning. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Definitions related to axioms? Compare the two ideas. Use a simple example.

22. How is axioms related to postulates? Compare the two ideas. Use a school-life example.

23. How is postulates related to deductive reasoning.? Compare the two ideas. Use a real-world example.

24. How is deductive reasoning. related to ■ Definitions? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Definitions related to axioms? Compare the two ideas. Explain in your own words.

26. How is axioms related to postulates? Compare the two ideas. Give a step-by-step response.

27. How is postulates related to deductive reasoning.? Compare the two ideas. Mention one common misconception.

28. How is deductive reasoning. related to ■ Definitions? Compare the two ideas. State one practical application.

29. How is ■ Definitions related to axioms? Compare the two ideas. Justify your response.

30. How is axioms related to postulates? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply postulates to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply deductive reasoning. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Definitions to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply axioms to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply postulates to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply deductive reasoning. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Definitions to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply axioms to a realistic situation. What would you do or calculate? State one practical application.

39. Apply postulates to a realistic situation. What would you do or calculate? Justify your response.

40. Apply deductive reasoning. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Definitions. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of axioms. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of postulates. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of deductive reasoning.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Definitions. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of axioms. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of postulates. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of deductive reasoning.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Definitions. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of axioms. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show postulates. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show deductive reasoning.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Definitions. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show axioms. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show postulates. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show deductive reasoning.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Definitions. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show axioms. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show postulates. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show deductive reasoning.. Include the relevant rule or principle.

MATHEMATICS — 6. Lines and Angles

Coverage: ■ Angle relationships; parallel lines; transversals; triangle angle properties.

Understanding

1. What is ■ Angle relationships, and what are its main features? Use a simple example.

2. What is parallel lines, and what are its main features? Use a school-life example.

3. What is transversals, and what are its main features? Use a real-world example.

4. What is triangle angle properties., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Angle relationships, and what are its main features? Explain in your own words.

6. What is parallel lines, and what are its main features? Give a step-by-step response.

7. What is transversals, and what are its main features? Mention one common misconception.

8. What is triangle angle properties., and what are its main features? State one practical application.

9. What is ■ Angle relationships, and what are its main features? Justify your response.

10. What is parallel lines, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to transversals and explain each. Use a simple example.

12. Give two examples related to triangle angle properties. and explain each. Use a school-life example.

13. Give two examples related to ■ Angle relationships and explain each. Use a real-world example.

14. Give two examples related to parallel lines and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to transversals and explain each. Explain in your own words.

16. Give two examples related to triangle angle properties. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Angle relationships and explain each. Mention one common misconception.

18. Give two examples related to parallel lines and explain each. State one practical application.

19. Give two examples related to transversals and explain each. Justify your response.

20. Give two examples related to triangle angle properties. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Angle relationships related to parallel lines? Compare the two ideas. Use a simple example.

22. How is parallel lines related to transversals? Compare the two ideas. Use a school-life example.

23. How is transversals related to triangle angle properties.? Compare the two ideas. Use a real-world example.

24. How is triangle angle properties. related to ■ Angle relationships? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Angle relationships related to parallel lines? Compare the two ideas. Explain in your own words.

26. How is parallel lines related to transversals? Compare the two ideas. Give a step-by-step response.

27. How is transversals related to triangle angle properties.? Compare the two ideas. Mention one common misconception.

28. How is triangle angle properties. related to ■ Angle relationships? Compare the two ideas. State one practical application.

29. How is ■ Angle relationships related to parallel lines? Compare the two ideas. Justify your response.

30. How is parallel lines related to transversals? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply transversals to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply triangle angle properties. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Angle relationships to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply parallel lines to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply transversals to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply triangle angle properties. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Angle relationships to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply parallel lines to a realistic situation. What would you do or calculate? State one practical application.

39. Apply transversals to a realistic situation. What would you do or calculate? Justify your response.

40. Apply triangle angle properties. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Angle relationships. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of parallel lines. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of transversals. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of triangle angle properties.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Angle relationships. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of parallel lines. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of transversals. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of triangle angle properties.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Angle relationships. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of parallel lines. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show transversals. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show triangle angle properties.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Angle relationships. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show parallel lines. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show transversals. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show triangle angle properties.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Angle relationships. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show parallel lines. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show transversals. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show triangle angle properties.. Include the relevant rule or principle.

MATHEMATICS — 7. Triangles

Coverage: ■ Congruence; properties; inequalities; proof-based reasoning.

Understanding

1. What is ■ Congruence, and what are its main features? Use a simple example.

2. What is properties, and what are its main features? Use a school-life example.

3. What is inequalities, and what are its main features? Use a real-world example.

4. What is proof-based reasoning., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Congruence, and what are its main features? Explain in your own words.

6. What is properties, and what are its main features? Give a step-by-step response.

7. What is inequalities, and what are its main features? Mention one common misconception.

8. What is proof-based reasoning., and what are its main features? State one practical application.

9. What is ■ Congruence, and what are its main features? Justify your response.

10. What is properties, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to inequalities and explain each. Use a simple example.

12. Give two examples related to proof-based reasoning. and explain each. Use a school-life example.

13. Give two examples related to ■ Congruence and explain each. Use a real-world example.

14. Give two examples related to properties and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to inequalities and explain each. Explain in your own words.

16. Give two examples related to proof-based reasoning. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Congruence and explain each. Mention one common misconception.

18. Give two examples related to properties and explain each. State one practical application.

19. Give two examples related to inequalities and explain each. Justify your response.

20. Give two examples related to proof-based reasoning. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Congruence related to properties? Compare the two ideas. Use a simple example.

22. How is properties related to inequalities? Compare the two ideas. Use a school-life example.

23. How is inequalities related to proof-based reasoning.? Compare the two ideas. Use a real-world example.

24. How is proof-based reasoning. related to ■ Congruence? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Congruence related to properties? Compare the two ideas. Explain in your own words.

26. How is properties related to inequalities? Compare the two ideas. Give a step-by-step response.

27. How is inequalities related to proof-based reasoning.? Compare the two ideas. Mention one common misconception.

28. How is proof-based reasoning. related to ■ Congruence? Compare the two ideas. State one practical application.

29. How is ■ Congruence related to properties? Compare the two ideas. Justify your response.

30. How is properties related to inequalities? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply inequalities to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply proof-based reasoning. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Congruence to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply properties to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply inequalities to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply proof-based reasoning. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Congruence to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply properties to a realistic situation. What would you do or calculate? State one practical application.

39. Apply inequalities to a realistic situation. What would you do or calculate? Justify your response.

40. Apply proof-based reasoning. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Congruence. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of properties. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of inequalities. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of proof-based reasoning.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Congruence. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of properties. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of inequalities. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of proof-based reasoning.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Congruence. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of properties. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show inequalities. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show proof-based reasoning.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Congruence. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show properties. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show inequalities. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show proof-based reasoning.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Congruence. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show properties. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show inequalities. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show proof-based reasoning.. Include the relevant rule or principle.

MATHEMATICS — 8. Quadrilaterals

Coverage: ■ Parallelograms and special quadrilaterals; diagonals; midpoint theorem.

Understanding

1. What is ■ Parallelograms and special quadrilaterals, and what are its main features? Use a simple example.

2. What is diagonals, and what are its main features? Use a school-life example.

3. What is midpoint theorem., and what are its main features? Use a real-world example.

4. What is ■ Parallelograms and special quadrilaterals, and what are its main features? Show the idea with a diagram where appropriate.

5. What is diagonals, and what are its main features? Explain in your own words.

6. What is midpoint theorem., and what are its main features? Give a step-by-step response.

7. What is ■ Parallelograms and special quadrilaterals, and what are its main features? Mention one common misconception.

8. What is diagonals, and what are its main features? State one practical application.

9. What is midpoint theorem., and what are its main features? Justify your response.

10. What is ■ Parallelograms and special quadrilaterals, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to diagonals and explain each. Use a simple example.

12. Give two examples related to midpoint theorem. and explain each. Use a school-life example.

13. Give two examples related to ■ Parallelograms and special quadrilaterals and explain each. Use a real-world example.

14. Give two examples related to diagonals and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to midpoint theorem. and explain each. Explain in your own words.

16. Give two examples related to ■ Parallelograms and special quadrilaterals and explain each. Give a step-by-step response.

17. Give two examples related to diagonals and explain each. Mention one common misconception.

18. Give two examples related to midpoint theorem. and explain each. State one practical application.

19. Give two examples related to ■ Parallelograms and special quadrilaterals and explain each. Justify your response.

20. Give two examples related to diagonals and explain each. Include the relevant rule or principle.

Comparison

21. How is midpoint theorem. related to ■ Parallelograms and special quadrilaterals? Compare the two ideas. Use a simple example.

22. How is ■ Parallelograms and special quadrilaterals related to diagonals? Compare the two ideas. Use a school-life example.

23. How is diagonals related to midpoint theorem.? Compare the two ideas. Use a real-world example.

24. How is midpoint theorem. related to ■ Parallelograms and special quadrilaterals? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Parallelograms and special quadrilaterals related to diagonals? Compare the two ideas. Explain in your own words.

26. How is diagonals related to midpoint theorem.? Compare the two ideas. Give a step-by-step response.

27. How is midpoint theorem. related to ■ Parallelograms and special quadrilaterals? Compare the two ideas. Mention one common misconception.

28. How is ■ Parallelograms and special quadrilaterals related to diagonals? Compare the two ideas. State one practical application.

29. How is diagonals related to midpoint theorem.? Compare the two ideas. Justify your response.

30. How is midpoint theorem. related to ■ Parallelograms and special quadrilaterals? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Parallelograms and special quadrilaterals to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply diagonals to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply midpoint theorem. to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply ■ Parallelograms and special quadrilaterals to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply diagonals to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply midpoint theorem. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Parallelograms and special quadrilaterals to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply diagonals to a realistic situation. What would you do or calculate? State one practical application.

39. Apply midpoint theorem. to a realistic situation. What would you do or calculate? Justify your response.

40. Apply ■ Parallelograms and special quadrilaterals to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of diagonals. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of midpoint theorem.. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of ■ Parallelograms and special quadrilaterals. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of diagonals. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of midpoint theorem.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Parallelograms and special quadrilaterals. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of diagonals. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of midpoint theorem.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Parallelograms and special quadrilaterals. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of diagonals. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show midpoint theorem.. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Parallelograms and special quadrilaterals. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show diagonals. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show midpoint theorem.. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Parallelograms and special quadrilaterals. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show diagonals. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show midpoint theorem.. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Parallelograms and special quadrilaterals. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show diagonals. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show midpoint theorem.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 1. Exploration: Entering the World of Secondary Science

Coverage: ■ Scientific inquiry; measurement; variables; evidence; interdisciplinary science.

Understanding

1. What is ■ Scientific inquiry, and what are its main features? Use a simple example.

2. What is measurement, and what are its main features? Use a school-life example.

3. What is variables, and what are its main features? Use a real-world example.

4. What is evidence, and what are its main features? Show the idea with a diagram where appropriate.

5. What is interdisciplinary science., and what are its main features? Explain in your own words.

6. What is ■ Scientific inquiry, and what are its main features? Give a step-by-step response.

7. What is measurement, and what are its main features? Mention one common misconception.

8. What is variables, and what are its main features? State one practical application.

9. What is evidence, and what are its main features? Justify your response.

10. What is interdisciplinary science., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Scientific inquiry and explain each. Use a simple example.

12. Give two examples related to measurement and explain each. Use a school-life example.

13. Give two examples related to variables and explain each. Use a real-world example.

14. Give two examples related to evidence and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to interdisciplinary science. and explain each. Explain in your own words.

16. Give two examples related to ■ Scientific inquiry and explain each. Give a step-by-step response.

17. Give two examples related to measurement and explain each. Mention one common misconception.

18. Give two examples related to variables and explain each. State one practical application.

19. Give two examples related to evidence and explain each. Justify your response.

20. Give two examples related to interdisciplinary science. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Scientific inquiry related to measurement? Compare the two ideas. Use a simple example.

22. How is measurement related to variables? Compare the two ideas. Use a school-life example.

23. How is variables related to evidence? Compare the two ideas. Use a real-world example.

24. How is evidence related to interdisciplinary science.? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is interdisciplinary science. related to ■ Scientific inquiry? Compare the two ideas. Explain in your own words.

26. How is ■ Scientific inquiry related to measurement? Compare the two ideas. Give a step-by-step response.

27. How is measurement related to variables? Compare the two ideas. Mention one common misconception.

28. How is variables related to evidence? Compare the two ideas. State one practical application.

29. How is evidence related to interdisciplinary science.? Compare the two ideas. Justify your response.

30. How is interdisciplinary science. related to ■ Scientific inquiry? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Scientific inquiry to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply measurement to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply variables to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply evidence to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply interdisciplinary science. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Scientific inquiry to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply measurement to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply variables to a realistic situation. What would you do or calculate? State one practical application.

39. Apply evidence to a realistic situation. What would you do or calculate? Justify your response.

40. Apply interdisciplinary science. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Scientific inquiry. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of measurement. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of variables. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of evidence. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of interdisciplinary science.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Scientific inquiry. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of measurement. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of variables. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of evidence. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of interdisciplinary science.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Scientific inquiry. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show measurement. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show variables. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show evidence. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show interdisciplinary science.. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Scientific inquiry. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show measurement. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show variables. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show evidence. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show interdisciplinary science.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 2. Cell: The Building Block of Life

Coverage: ■ Cell theory; cell structure; organelles; microscopy; plant/animal cells.

Understanding

1. What is ■ Cell theory, and what are its main features? Use a simple example.

2. What is cell structure, and what are its main features? Use a school-life example.

3. What is organelles, and what are its main features? Use a real-world example.

4. What is microscopy, and what are its main features? Show the idea with a diagram where appropriate.

5. What is plant/animal cells., and what are its main features? Explain in your own words.

6. What is ■ Cell theory, and what are its main features? Give a step-by-step response.

7. What is cell structure, and what are its main features? Mention one common misconception.

8. What is organelles, and what are its main features? State one practical application.

9. What is microscopy, and what are its main features? Justify your response.

10. What is plant/animal cells., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Cell theory and explain each. Use a simple example.

12. Give two examples related to cell structure and explain each. Use a school-life example.

13. Give two examples related to organelles and explain each. Use a real-world example.

14. Give two examples related to microscopy and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to plant/animal cells. and explain each. Explain in your own words.

16. Give two examples related to ■ Cell theory and explain each. Give a step-by-step response.

17. Give two examples related to cell structure and explain each. Mention one common misconception.

18. Give two examples related to organelles and explain each. State one practical application.

19. Give two examples related to microscopy and explain each. Justify your response.

20. Give two examples related to plant/animal cells. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Cell theory related to cell structure? Compare the two ideas. Use a simple example.

22. How is cell structure related to organelles? Compare the two ideas. Use a school-life example.

23. How is organelles related to microscopy? Compare the two ideas. Use a real-world example.

24. How is microscopy related to plant/animal cells.? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is plant/animal cells. related to ■ Cell theory? Compare the two ideas. Explain in your own words.

26. How is ■ Cell theory related to cell structure? Compare the two ideas. Give a step-by-step response.

27. How is cell structure related to organelles? Compare the two ideas. Mention one common misconception.

28. How is organelles related to microscopy? Compare the two ideas. State one practical application.

29. How is microscopy related to plant/animal cells.? Compare the two ideas. Justify your response.

30. How is plant/animal cells. related to ■ Cell theory? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Cell theory to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply cell structure to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply organelles to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply microscopy to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply plant/animal cells. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Cell theory to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply cell structure to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply organelles to a realistic situation. What would you do or calculate? State one practical application.

39. Apply microscopy to a realistic situation. What would you do or calculate? Justify your response.

40. Apply plant/animal cells. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Cell theory. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of cell structure. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of organelles. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of microscopy. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of plant/animal cells.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Cell theory. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of cell structure. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of organelles. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of microscopy. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of plant/animal cells.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Cell theory. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show cell structure. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show organelles. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show microscopy. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show plant/animal cells.. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Cell theory. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show cell structure. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show organelles. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show microscopy. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show plant/animal cells.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 3. Tissues in Action

Coverage: ■ Plant and animal tissues; structure-function relationships; organisation.

Understanding

1. What is ■ Plant and animal tissues, and what are its main features? Use a simple example.

2. What is structure-function relationships, and what are its main features? Use a school-life example.

3. What is organisation., and what are its main features? Use a real-world example.

4. What is ■ Plant and animal tissues, and what are its main features? Show the idea with a diagram where appropriate.

5. What is structure-function relationships, and what are its main features? Explain in your own words.

6. What is organisation., and what are its main features? Give a step-by-step response.

7. What is ■ Plant and animal tissues, and what are its main features? Mention one common misconception.

8. What is structure-function relationships, and what are its main features? State one practical application.

9. What is organisation., and what are its main features? Justify your response.

10. What is ■ Plant and animal tissues, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to structure-function relationships and explain each. Use a simple example.

12. Give two examples related to organisation. and explain each. Use a school-life example.

13. Give two examples related to ■ Plant and animal tissues and explain each. Use a real-world example.

14. Give two examples related to structure-function relationships and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to organisation. and explain each. Explain in your own words.

16. Give two examples related to ■ Plant and animal tissues and explain each. Give a step-by-step response.

17. Give two examples related to structure-function relationships and explain each. Mention one common misconception.

18. Give two examples related to organisation. and explain each. State one practical application.

19. Give two examples related to ■ Plant and animal tissues and explain each. Justify your response.

20. Give two examples related to structure-function relationships and explain each. Include the relevant rule or principle.

Comparison

21. How is organisation. related to ■ Plant and animal tissues? Compare the two ideas. Use a simple example.

22. How is ■ Plant and animal tissues related to structure-function relationships? Compare the two ideas. Use a school-life example.

23. How is structure-function relationships related to organisation.? Compare the two ideas. Use a real-world example.

24. How is organisation. related to ■ Plant and animal tissues? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Plant and animal tissues related to structure-function relationships? Compare the two ideas. Explain in your own words.

26. How is structure-function relationships related to organisation.? Compare the two ideas. Give a step-by-step response.

27. How is organisation. related to ■ Plant and animal tissues? Compare the two ideas. Mention one common misconception.

28. How is ■ Plant and animal tissues related to structure-function relationships? Compare the two ideas. State one practical application.

29. How is structure-function relationships related to organisation.? Compare the two ideas. Justify your response.

30. How is organisation. related to ■ Plant and animal tissues? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Plant and animal tissues to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply structure-function relationships to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply organisation. to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply ■ Plant and animal tissues to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply structure-function relationships to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply organisation. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Plant and animal tissues to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply structure-function relationships to a realistic situation. What would you do or calculate? State one practical application.

39. Apply organisation. to a realistic situation. What would you do or calculate? Justify your response.

40. Apply ■ Plant and animal tissues to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of structure-function relationships. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of organisation.. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of ■ Plant and animal tissues. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of structure-function relationships. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of organisation.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Plant and animal tissues. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of structure-function relationships. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of organisation.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Plant and animal tissues. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of structure-function relationships. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show organisation.. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Plant and animal tissues. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show structure-function relationships. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show organisation.. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Plant and animal tissues. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show structure-function relationships. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show organisation.. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Plant and animal tissues. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show structure-function relationships. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show organisation.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 4. Describing Motion Around Us

Coverage: ■ Distance/displacement; speed/velocity; acceleration; graphs.

Understanding

1. What is ■ Distance/displacement, and what are its main features? Use a simple example.

2. What is speed/velocity, and what are its main features? Use a school-life example.

3. What is acceleration, and what are its main features? Use a real-world example.

4. What is graphs., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Distance/displacement, and what are its main features? Explain in your own words.

6. What is speed/velocity, and what are its main features? Give a step-by-step response.

7. What is acceleration, and what are its main features? Mention one common misconception.

8. What is graphs., and what are its main features? State one practical application.

9. What is ■ Distance/displacement, and what are its main features? Justify your response.

10. What is speed/velocity, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to acceleration and explain each. Use a simple example.

12. Give two examples related to graphs. and explain each. Use a school-life example.

13. Give two examples related to ■ Distance/displacement and explain each. Use a real-world example.

14. Give two examples related to speed/velocity and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to acceleration and explain each. Explain in your own words.

16. Give two examples related to graphs. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Distance/displacement and explain each. Mention one common misconception.

18. Give two examples related to speed/velocity and explain each. State one practical application.

19. Give two examples related to acceleration and explain each. Justify your response.

20. Give two examples related to graphs. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Distance/displacement related to speed/velocity? Compare the two ideas. Use a simple example.

22. How is speed/velocity related to acceleration? Compare the two ideas. Use a school-life example.

23. How is acceleration related to graphs.? Compare the two ideas. Use a real-world example.

24. How is graphs. related to ■ Distance/displacement? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Distance/displacement related to speed/velocity? Compare the two ideas. Explain in your own words.

26. How is speed/velocity related to acceleration? Compare the two ideas. Give a step-by-step response.

27. How is acceleration related to graphs.? Compare the two ideas. Mention one common misconception.

28. How is graphs. related to ■ Distance/displacement? Compare the two ideas. State one practical application.

29. How is ■ Distance/displacement related to speed/velocity? Compare the two ideas. Justify your response.

30. How is speed/velocity related to acceleration? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply acceleration to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply graphs. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Distance/displacement to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply speed/velocity to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply acceleration to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply graphs. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Distance/displacement to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply speed/velocity to a realistic situation. What would you do or calculate? State one practical application.

39. Apply acceleration to a realistic situation. What would you do or calculate? Justify your response.

40. Apply graphs. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Distance/displacement. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of speed/velocity. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of acceleration. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of graphs.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Distance/displacement. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of speed/velocity. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of acceleration. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of graphs.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Distance/displacement. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of speed/velocity. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show acceleration. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show graphs.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distance/displacement. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show speed/velocity. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show acceleration. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show graphs.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distance/displacement. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show speed/velocity. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show acceleration. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show graphs.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 5. Exploring Mixtures and their Separation

Coverage: ■ Mixtures; solutions; separation methods; physical properties.

Understanding

1. What is ■ Mixtures, and what are its main features? Use a simple example.

2. What is solutions, and what are its main features? Use a school-life example.

3. What is separation methods, and what are its main features? Use a real-world example.

4. What is physical properties., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Mixtures, and what are its main features? Explain in your own words.

6. What is solutions, and what are its main features? Give a step-by-step response.

7. What is separation methods, and what are its main features? Mention one common misconception.

8. What is physical properties., and what are its main features? State one practical application.

9. What is ■ Mixtures, and what are its main features? Justify your response.

10. What is solutions, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to separation methods and explain each. Use a simple example.

12. Give two examples related to physical properties. and explain each. Use a school-life example.

13. Give two examples related to ■ Mixtures and explain each. Use a real-world example.

14. Give two examples related to solutions and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to separation methods and explain each. Explain in your own words.

16. Give two examples related to physical properties. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Mixtures and explain each. Mention one common misconception.

18. Give two examples related to solutions and explain each. State one practical application.

19. Give two examples related to separation methods and explain each. Justify your response.

20. Give two examples related to physical properties. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Mixtures related to solutions? Compare the two ideas. Use a simple example.

22. How is solutions related to separation methods? Compare the two ideas. Use a school-life example.

23. How is separation methods related to physical properties.? Compare the two ideas. Use a real-world example.

24. How is physical properties. related to ■ Mixtures? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Mixtures related to solutions? Compare the two ideas. Explain in your own words.

26. How is solutions related to separation methods? Compare the two ideas. Give a step-by-step response.

27. How is separation methods related to physical properties.? Compare the two ideas. Mention one common misconception.

28. How is physical properties. related to ■ Mixtures? Compare the two ideas. State one practical application.

29. How is ■ Mixtures related to solutions? Compare the two ideas. Justify your response.

30. How is solutions related to separation methods? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply separation methods to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply physical properties. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Mixtures to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply solutions to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply separation methods to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply physical properties. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Mixtures to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply solutions to a realistic situation. What would you do or calculate? State one practical application.

39. Apply separation methods to a realistic situation. What would you do or calculate? Justify your response.

40. Apply physical properties. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Mixtures. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of solutions. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of separation methods. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of physical properties.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Mixtures. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of solutions. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of separation methods. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of physical properties.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Mixtures. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of solutions. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show separation methods. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show physical properties.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Mixtures. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show solutions. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show separation methods. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show physical properties.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Mixtures. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show solutions. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show separation methods. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show physical properties.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 6. How Forces Affect Motion

Coverage: ■ Force; inertia; acceleration; Newtonian ideas; balanced/unbalanced forces.

Understanding

1. What is ■ Force, and what are its main features? Use a simple example.

2. What is inertia, and what are its main features? Use a school-life example.

3. What is acceleration, and what are its main features? Use a real-world example.

4. What is Newtonian ideas, and what are its main features? Show the idea with a diagram where appropriate.

5. What is balanced/unbalanced forces., and what are its main features? Explain in your own words.

6. What is ■ Force, and what are its main features? Give a step-by-step response.

7. What is inertia, and what are its main features? Mention one common misconception.

8. What is acceleration, and what are its main features? State one practical application.

9. What is Newtonian ideas, and what are its main features? Justify your response.

10. What is balanced/unbalanced forces., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Force and explain each. Use a simple example.

12. Give two examples related to inertia and explain each. Use a school-life example.

13. Give two examples related to acceleration and explain each. Use a real-world example.

14. Give two examples related to Newtonian ideas and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to balanced/unbalanced forces. and explain each. Explain in your own words.

16. Give two examples related to ■ Force and explain each. Give a step-by-step response.

17. Give two examples related to inertia and explain each. Mention one common misconception.

18. Give two examples related to acceleration and explain each. State one practical application.

19. Give two examples related to Newtonian ideas and explain each. Justify your response.

20. Give two examples related to balanced/unbalanced forces. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Force related to inertia? Compare the two ideas. Use a simple example.

22. How is inertia related to acceleration? Compare the two ideas. Use a school-life example.

23. How is acceleration related to Newtonian ideas? Compare the two ideas. Use a real-world example.

24. How is Newtonian ideas related to balanced/unbalanced forces.? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is balanced/unbalanced forces. related to ■ Force? Compare the two ideas. Explain in your own words.

26. How is ■ Force related to inertia? Compare the two ideas. Give a step-by-step response.

27. How is inertia related to acceleration? Compare the two ideas. Mention one common misconception.

28. How is acceleration related to Newtonian ideas? Compare the two ideas. State one practical application.

29. How is Newtonian ideas related to balanced/unbalanced forces.? Compare the two ideas. Justify your response.

30. How is balanced/unbalanced forces. related to ■ Force? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Force to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply inertia to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply acceleration to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply Newtonian ideas to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply balanced/unbalanced forces. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Force to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply inertia to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply acceleration to a realistic situation. What would you do or calculate? State one practical application.

39. Apply Newtonian ideas to a realistic situation. What would you do or calculate? Justify your response.

40. Apply balanced/unbalanced forces. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Force. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of inertia. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of acceleration. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of Newtonian ideas. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of balanced/unbalanced forces.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Force. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of inertia. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of acceleration. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of Newtonian ideas. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of balanced/unbalanced forces.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Force. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show inertia. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show acceleration. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show Newtonian ideas. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show balanced/unbalanced forces.. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Force. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show inertia. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show acceleration. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show Newtonian ideas. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show balanced/unbalanced forces.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 7. Work, Energy, and Simple Machines

Coverage: ■ Work; energy; power; simple machines; efficiency and applications.

Understanding

1. What is ■ Work, and what are its main features? Use a simple example.

2. What is energy, and what are its main features? Use a school-life example.

3. What is power, and what are its main features? Use a real-world example.

4. What is simple machines, and what are its main features? Show the idea with a diagram where appropriate.

5. What is efficiency and applications., and what are its main features? Explain in your own words.

6. What is ■ Work, and what are its main features? Give a step-by-step response.

7. What is energy, and what are its main features? Mention one common misconception.

8. What is power, and what are its main features? State one practical application.

9. What is simple machines, and what are its main features? Justify your response.

10. What is efficiency and applications., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Work and explain each. Use a simple example.

12. Give two examples related to energy and explain each. Use a school-life example.

13. Give two examples related to power and explain each. Use a real-world example.

14. Give two examples related to simple machines and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to efficiency and applications. and explain each. Explain in your own words.

16. Give two examples related to ■ Work and explain each. Give a step-by-step response.

17. Give two examples related to energy and explain each. Mention one common misconception.

18. Give two examples related to power and explain each. State one practical application.

19. Give two examples related to simple machines and explain each. Justify your response.

20. Give two examples related to efficiency and applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Work related to energy? Compare the two ideas. Use a simple example.

22. How is energy related to power? Compare the two ideas. Use a school-life example.

23. How is power related to simple machines? Compare the two ideas. Use a real-world example.

24. How is simple machines related to efficiency and applications.? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is efficiency and applications. related to ■ Work? Compare the two ideas. Explain in your own words.

26. How is ■ Work related to energy? Compare the two ideas. Give a step-by-step response.

27. How is energy related to power? Compare the two ideas. Mention one common misconception.

28. How is power related to simple machines? Compare the two ideas. State one practical application.

29. How is simple machines related to efficiency and applications.? Compare the two ideas. Justify your response.

30. How is efficiency and applications. related to ■ Work? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Work to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply energy to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply power to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply simple machines to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply efficiency and applications. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Work to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply energy to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply power to a realistic situation. What would you do or calculate? State one practical application.

39. Apply simple machines to a realistic situation. What would you do or calculate? Justify your response.

40. Apply efficiency and applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Work. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of energy. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of power. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of simple machines. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of efficiency and applications.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Work. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of energy. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of power. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of simple machines. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of efficiency and applications.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Work. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show energy. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show power. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show simple machines. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show efficiency and applications.. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Work. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show energy. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show power. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show simple machines. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show efficiency and applications.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 8. Journey Inside the Atom

Coverage: ■ Subatomic particles; atomic structure; models; electron arrangement.

Understanding

1. What is ■ Subatomic particles, and what are its main features? Use a simple example.

2. What is atomic structure, and what are its main features? Use a school-life example.

3. What is models, and what are its main features? Use a real-world example.

4. What is electron arrangement., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Subatomic particles, and what are its main features? Explain in your own words.

6. What is atomic structure, and what are its main features? Give a step-by-step response.

7. What is models, and what are its main features? Mention one common misconception.

8. What is electron arrangement., and what are its main features? State one practical application.

9. What is ■ Subatomic particles, and what are its main features? Justify your response.

10. What is atomic structure, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to models and explain each. Use a simple example.

12. Give two examples related to electron arrangement. and explain each. Use a school-life example.

13. Give two examples related to ■ Subatomic particles and explain each. Use a real-world example.

14. Give two examples related to atomic structure and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to models and explain each. Explain in your own words.

16. Give two examples related to electron arrangement. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Subatomic particles and explain each. Mention one common misconception.

18. Give two examples related to atomic structure and explain each. State one practical application.

19. Give two examples related to models and explain each. Justify your response.

20. Give two examples related to electron arrangement. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Subatomic particles related to atomic structure? Compare the two ideas. Use a simple example.

22. How is atomic structure related to models? Compare the two ideas. Use a school-life example.

23. How is models related to electron arrangement.? Compare the two ideas. Use a real-world example.

24. How is electron arrangement. related to ■ Subatomic particles? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Subatomic particles related to atomic structure? Compare the two ideas. Explain in your own words.

26. How is atomic structure related to models? Compare the two ideas. Give a step-by-step response.

27. How is models related to electron arrangement.? Compare the two ideas. Mention one common misconception.

28. How is electron arrangement. related to ■ Subatomic particles? Compare the two ideas. State one practical application.

29. How is ■ Subatomic particles related to atomic structure? Compare the two ideas. Justify your response.

30. How is atomic structure related to models? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply models to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply electron arrangement. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Subatomic particles to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply atomic structure to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply models to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply electron arrangement. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Subatomic particles to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply atomic structure to a realistic situation. What would you do or calculate? State one practical application.

39. Apply models to a realistic situation. What would you do or calculate? Justify your response.

40. Apply electron arrangement. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Subatomic particles. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of atomic structure. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of models. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of electron arrangement.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Subatomic particles. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of atomic structure. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of models. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of electron arrangement.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Subatomic particles. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of atomic structure. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show models. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show electron arrangement.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Subatomic particles. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show atomic structure. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show models. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show electron arrangement.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Subatomic particles. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show atomic structure. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show models. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show electron arrangement.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 9. Atomic Foundations of Matter

Coverage: ■ Atoms, molecules, elements and compounds; chemical representations.

Understanding

1. What is ■ Atoms, and what are its main features? Use a simple example.

2. What is molecules, and what are its main features? Use a school-life example.

3. What is elements and compounds, and what are its main features? Use a real-world example.

4. What is chemical representations., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Atoms, and what are its main features? Explain in your own words.

6. What is molecules, and what are its main features? Give a step-by-step response.

7. What is elements and compounds, and what are its main features? Mention one common misconception.

8. What is chemical representations., and what are its main features? State one practical application.

9. What is ■ Atoms, and what are its main features? Justify your response.

10. What is molecules, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to elements and compounds and explain each. Use a simple example.

12. Give two examples related to chemical representations. and explain each. Use a school-life example.

13. Give two examples related to ■ Atoms and explain each. Use a real-world example.

14. Give two examples related to molecules and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to elements and compounds and explain each. Explain in your own words.

16. Give two examples related to chemical representations. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Atoms and explain each. Mention one common misconception.

18. Give two examples related to molecules and explain each. State one practical application.

19. Give two examples related to elements and compounds and explain each. Justify your response.

20. Give two examples related to chemical representations. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Atoms related to molecules? Compare the two ideas. Use a simple example.

22. How is molecules related to elements and compounds? Compare the two ideas. Use a school-life example.

23. How is elements and compounds related to chemical representations.? Compare the two ideas. Use a real-world example.

24. How is chemical representations. related to ■ Atoms? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Atoms related to molecules? Compare the two ideas. Explain in your own words.

26. How is molecules related to elements and compounds? Compare the two ideas. Give a step-by-step response.

27. How is elements and compounds related to chemical representations.? Compare the two ideas. Mention one common misconception.

28. How is chemical representations. related to ■ Atoms? Compare the two ideas. State one practical application.

29. How is ■ Atoms related to molecules? Compare the two ideas. Justify your response.

30. How is molecules related to elements and compounds? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply elements and compounds to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply chemical representations. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Atoms to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply molecules to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply elements and compounds to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply chemical representations. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Atoms to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply molecules to a realistic situation. What would you do or calculate? State one practical application.

39. Apply elements and compounds to a realistic situation. What would you do or calculate? Justify your response.

40. Apply chemical representations. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Atoms. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of molecules. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of elements and compounds. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of chemical representations.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Atoms. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of molecules. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of elements and compounds. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of chemical representations.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Atoms. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of molecules. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show elements and compounds. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show chemical representations.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Atoms. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show molecules. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show elements and compounds. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show chemical representations.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Atoms. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show molecules. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show elements and compounds. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show chemical representations.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 10. Sound Waves: Characteristics and Applications

Coverage: NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus

Understanding

1. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a simple example.

2. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a school-life example.

3. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a real-world example.

4. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Show the idea with a diagram where appropriate.

5. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Explain in your own words.

6. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Give a step-by-step response.

7. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Mention one common misconception.

8. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? State one practical application.

9. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Justify your response.

10. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a simple example.

12. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a school-life example.

13. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a real-world example.

14. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Explain in your own words.

16. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Give a step-by-step response.

17. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Mention one common misconception.

18. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. State one practical application.

19. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Justify your response.

20. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Include the relevant rule or principle.

Comparison

21. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a simple example.

22. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a school-life example.

23. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a real-world example.

24. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Explain in your own words.

26. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Give a step-by-step response.

27. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Mention one common misconception.

28. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. State one practical application.

29. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Justify your response.

30. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? State one practical application.

39. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Justify your response.

40. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 11. Reproduction: How Life Continues

Coverage: ■ Asexual/sexual reproduction; reproductive structures; continuity of life.

Understanding

1. What is ■ Asexual/sexual reproduction, and what are its main features? Use a simple example.

2. What is reproductive structures, and what are its main features? Use a school-life example.

3. What is continuity of life., and what are its main features? Use a real-world example.

4. What is ■ Asexual/sexual reproduction, and what are its main features? Show the idea with a diagram where appropriate.

5. What is reproductive structures, and what are its main features? Explain in your own words.

6. What is continuity of life., and what are its main features? Give a step-by-step response.

7. What is ■ Asexual/sexual reproduction, and what are its main features? Mention one common misconception.

8. What is reproductive structures, and what are its main features? State one practical application.

9. What is continuity of life., and what are its main features? Justify your response.

10. What is ■ Asexual/sexual reproduction, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to reproductive structures and explain each. Use a simple example.

12. Give two examples related to continuity of life. and explain each. Use a school-life example.

13. Give two examples related to ■ Asexual/sexual reproduction and explain each. Use a real-world example.

14. Give two examples related to reproductive structures and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to continuity of life. and explain each. Explain in your own words.

16. Give two examples related to ■ Asexual/sexual reproduction and explain each. Give a step-by-step response.

17. Give two examples related to reproductive structures and explain each. Mention one common misconception.

18. Give two examples related to continuity of life. and explain each. State one practical application.

19. Give two examples related to ■ Asexual/sexual reproduction and explain each. Justify your response.

20. Give two examples related to reproductive structures and explain each. Include the relevant rule or principle.

Comparison

21. How is continuity of life. related to ■ Asexual/sexual reproduction? Compare the two ideas. Use a simple example.

22. How is ■ Asexual/sexual reproduction related to reproductive structures? Compare the two ideas. Use a school-life example.

23. How is reproductive structures related to continuity of life.? Compare the two ideas. Use a real-world example.

24. How is continuity of life. related to ■ Asexual/sexual reproduction? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Asexual/sexual reproduction related to reproductive structures? Compare the two ideas. Explain in your own words.

26. How is reproductive structures related to continuity of life.? Compare the two ideas. Give a step-by-step response.

27. How is continuity of life. related to ■ Asexual/sexual reproduction? Compare the two ideas. Mention one common misconception.

28. How is ■ Asexual/sexual reproduction related to reproductive structures? Compare the two ideas. State one practical application.

29. How is reproductive structures related to continuity of life.? Compare the two ideas. Justify your response.

30. How is continuity of life. related to ■ Asexual/sexual reproduction? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Asexual/sexual reproduction to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply reproductive structures to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply continuity of life. to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply ■ Asexual/sexual reproduction to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply reproductive structures to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply continuity of life. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Asexual/sexual reproduction to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply reproductive structures to a realistic situation. What would you do or calculate? State one practical application.

39. Apply continuity of life. to a realistic situation. What would you do or calculate? Justify your response.

40. Apply ■ Asexual/sexual reproduction to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of reproductive structures. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of continuity of life.. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of ■ Asexual/sexual reproduction. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of reproductive structures. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of continuity of life.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Asexual/sexual reproduction. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of reproductive structures. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of continuity of life.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Asexual/sexual reproduction. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of reproductive structures. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show continuity of life.. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Asexual/sexual reproduction. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive structures. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show continuity of life.. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Asexual/sexual reproduction. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive structures. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show continuity of life.. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Asexual/sexual reproduction. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive structures. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show continuity of life.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 12. Patterns in Life: Diversity and Classification

Coverage: ■ Classification; biodiversity; taxonomic patterns; evolutionary relationships.

Understanding

1. What is ■ Classification, and what are its main features? Use a simple example.

2. What is biodiversity, and what are its main features? Use a school-life example.

3. What is taxonomic patterns, and what are its main features? Use a real-world example.

4. What is evolutionary relationships., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Classification, and what are its main features? Explain in your own words.

6. What is biodiversity, and what are its main features? Give a step-by-step response.

7. What is taxonomic patterns, and what are its main features? Mention one common misconception.

8. What is evolutionary relationships., and what are its main features? State one practical application.

9. What is ■ Classification, and what are its main features? Justify your response.

10. What is biodiversity, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to taxonomic patterns and explain each. Use a simple example.

12. Give two examples related to evolutionary relationships. and explain each. Use a school-life example.

13. Give two examples related to ■ Classification and explain each. Use a real-world example.

14. Give two examples related to biodiversity and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to taxonomic patterns and explain each. Explain in your own words.

16. Give two examples related to evolutionary relationships. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Classification and explain each. Mention one common misconception.

18. Give two examples related to biodiversity and explain each. State one practical application.

19. Give two examples related to taxonomic patterns and explain each. Justify your response.

20. Give two examples related to evolutionary relationships. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Classification related to biodiversity? Compare the two ideas. Use a simple example.

22. How is biodiversity related to taxonomic patterns? Compare the two ideas. Use a school-life example.

23. How is taxonomic patterns related to evolutionary relationships.? Compare the two ideas. Use a real-world example.

24. How is evolutionary relationships. related to ■ Classification? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Classification related to biodiversity? Compare the two ideas. Explain in your own words.

26. How is biodiversity related to taxonomic patterns? Compare the two ideas. Give a step-by-step response.

27. How is taxonomic patterns related to evolutionary relationships.? Compare the two ideas. Mention one common misconception.

28. How is evolutionary relationships. related to ■ Classification? Compare the two ideas. State one practical application.

29. How is ■ Classification related to biodiversity? Compare the two ideas. Justify your response.

30. How is biodiversity related to taxonomic patterns? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply taxonomic patterns to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply evolutionary relationships. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Classification to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply biodiversity to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply taxonomic patterns to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply evolutionary relationships. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Classification to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply biodiversity to a realistic situation. What would you do or calculate? State one practical application.

39. Apply taxonomic patterns to a realistic situation. What would you do or calculate? Justify your response.

40. Apply evolutionary relationships. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Classification. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of biodiversity. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of taxonomic patterns. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of evolutionary relationships.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Classification. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of biodiversity. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of taxonomic patterns. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of evolutionary relationships.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Classification. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of biodiversity. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show taxonomic patterns. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show evolutionary relationships.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Classification. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show biodiversity. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show taxonomic patterns. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show evolutionary relationships.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Classification. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show biodiversity. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show taxonomic patterns. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show evolutionary relationships.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 13. Earth as a System: Energy, Matter, and Life

Coverage: ■ Earth systems; energy/material cycles; atmosphere; biosphere; human impact.

Understanding

1. What is ■ Earth systems, and what are its main features? Use a simple example.

2. What is energy/material cycles, and what are its main features? Use a school-life example.

3. What is atmosphere, and what are its main features? Use a real-world example.

4. What is biosphere, and what are its main features? Show the idea with a diagram where appropriate.

5. What is human impact., and what are its main features? Explain in your own words.

6. What is ■ Earth systems, and what are its main features? Give a step-by-step response.

7. What is energy/material cycles, and what are its main features? Mention one common misconception.

8. What is atmosphere, and what are its main features? State one practical application.

9. What is biosphere, and what are its main features? Justify your response.

10. What is human impact., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Earth systems and explain each. Use a simple example.

12. Give two examples related to energy/material cycles and explain each. Use a school-life example.

13. Give two examples related to atmosphere and explain each. Use a real-world example.

14. Give two examples related to biosphere and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to human impact. and explain each. Explain in your own words.

16. Give two examples related to ■ Earth systems and explain each. Give a step-by-step response.

17. Give two examples related to energy/material cycles and explain each. Mention one common misconception.

18. Give two examples related to atmosphere and explain each. State one practical application.

19. Give two examples related to biosphere and explain each. Justify your response.

20. Give two examples related to human impact. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Earth systems related to energy/material cycles? Compare the two ideas. Use a simple example.

22. How is energy/material cycles related to atmosphere? Compare the two ideas. Use a school-life example.

23. How is atmosphere related to biosphere? Compare the two ideas. Use a real-world example.

24. How is biosphere related to human impact.? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is human impact. related to ■ Earth systems? Compare the two ideas. Explain in your own words.

26. How is ■ Earth systems related to energy/material cycles? Compare the two ideas. Give a step-by-step response.

27. How is energy/material cycles related to atmosphere? Compare the two ideas. Mention one common misconception.

28. How is atmosphere related to biosphere? Compare the two ideas. State one practical application.

29. How is biosphere related to human impact.? Compare the two ideas. Justify your response.

30. How is human impact. related to ■ Earth systems? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Earth systems to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply energy/material cycles to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply atmosphere to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply biosphere to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply human impact. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Earth systems to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply energy/material cycles to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply atmosphere to a realistic situation. What would you do or calculate? State one practical application.

39. Apply biosphere to a realistic situation. What would you do or calculate? Justify your response.

40. Apply human impact. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Earth systems. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of energy/material cycles. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of atmosphere. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of biosphere. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of human impact.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Earth systems. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of energy/material cycles. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of atmosphere. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of biosphere. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of human impact.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Earth systems. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show energy/material cycles. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show atmosphere. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show biosphere. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show human impact.. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Earth systems. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show energy/material cycles. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show atmosphere. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show biosphere. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show human impact.. Include the relevant rule or principle.